

EQF-001 — Epistemic Qualification Fabric

Status: FROZEN — MIP-001 W0 / W4 **Constitutional basis:** Articles 1, 3, 11, 28, 29, 30 **Executable form:**
pkg/qualification/{reverseproof,observability,independence,state,nextbest}

1. Purpose

EQF is where VERIQO's actual intellectual claim lives. Anyone can hash a file. EQF is the machinery that asks, for a given claim:

What must be true? What should we observe? What would contradict it? What alternative explains the same observations? What is missing? What could we not observe? What would most reduce uncertainty?

and then refuses to let a finding outrun the answers.

2. Core pipeline

Claim → Reverse Proof → Expected Conditions → Evidence Requirements
→ Observations → Alternative Hypotheses → Contradiction Matrix
→ Trust Calculus → Dissent → Qualification

3. Reverse proof (W4.1)

For every material claim, reverse proof derives an **EvidenceRequirementSet** by asking the seven questions above. The output is not a score — it is a *list of things that would settle the question*, each tagged with whether it was obtained, could not be obtained, or was never attempted.

This inverts the ordinary analytics posture. Instead of "here is what the data shows," EQF starts from "here is what the world would look like if the claim were true," then measures the gap.

4. Observed absence (W4.2)

Absence is never automatically evidence. Seven distinct states:

State	Meaning
OBSERVED_ABSENT	We looked, with adequate coverage, and it was not there.
EXPECTED_BUT_NOT_TESTED	We expected it and did not look.
NOT_OBSERVABLE	No source could have shown it.
NOT_COLLECTABLE	Observable in principle, not acquirable by us.
PARTIAL_COVERAGE	We looked, but coverage was incomplete.
SOURCE_UNAVAILABLE	The source existed but was down/unreachable.
INCONCLUSIVE	We looked and cannot say.

OBSERVED_ABSENT — the only one that carries evidential weight — requires **all nine** of: adequate source, operational availability, known coverage, valid query, valid expectation, correct temporal scope, correct spatial scope, integrity, and review where material.

The observability gate enforces this. A caller asserting **OBSERVED_ABSENT** without the nine conditions gets a refusal, not a warning (Article 29).

This is the difference between *"the vessel did not transmit AIS"* and *"we were not receiving AIS in that region during that window."* The first is a finding. The second is an infrastructure fact about us.

5. Source independence (W4.3)

Fifteen dimensions, each independently assessed:

RootOrigin · SensorMeasurement · ProviderPipeline · Collector
OrganizationalControl · AcquisitionPath · TransformationLineage
Temporal · Spatial · Methodological · Incentive · IdentityAssurance
DataCustody · ModelDependency · LegalMandate

Two rules govern the result:

1. **Same-root data is one source** (Article 3). Two feeds that both originate from the same satellite constellation, the same provider pipeline, or the same collector do not corroborate each other, however different their file formats.
2. **UNKNOWN is not INDEPENDENT** (Article 28). An unassessed dependency dimension leaves the pair **UNKNOWN**. It never defaults to independent, and it never quietly becomes independent by being ignored.

The second rule is the one that costs something. It means a great many real corroboration claims degrade to **UNKNOWN** — which is the correct and honest outcome.

6. Contradiction matrix (W4.4)

Every material hypothesis is scored against evidence, observations, expected observations, counterfactuals, missing evidence, and observed absence. Eight outcomes:

SUPPORTS · WEAKENS · CONTRADICTS · NEUTRAL
MISSING · NOT_OBSERVABLE · OBSERVED_ABSENT · INCONCLUSIVE

The last four are not "no data" — they are substantive states with different epistemic weight, and collapsing them into a single null is a violation of Article 30.

7. Trust calculus (W20)

Nine dimensions: source reliability, acquisition integrity, acquisition independence, evidentiary independence, observability quality, relevance, freshness, contradiction state, calibration.

Independence is not added as a confidence number. It is a gate: an unknown dependency keeps the assessment at **UNKNOWN** rather than contributing a discounted score. A weighted average that silently absorbs an unknown dependency into a 0.7 confidence is exactly the failure Article 28 exists to prevent.

8. Qualification states (W4.5)

SUPPORTED · SUPPORTED_WITH_EXCEPTIONS · QUALIFIED
QUALIFIED_WITH_DISSENT · INCONCLUSIVE · INSUFFICIENT_EVIDENCE
CONTRADICTED · NOT_OBSERVABLE · NOT_COLLECTABLE

No state maps to a legal conclusion (Article 16). There is deliberately no `PROVEN` and no `ESTABLISHED`.

9. Single-source exception (§29)

Where independent corroboration is genuinely impossible, a `SingleSourceException` may be recorded — with why single source is necessary, why alternatives are unavailable, source assurance, coverage, known limitations, reviewer, policy, and expiry.

Its output state is:

`SUPPORTED_BY_SINGLE_HIGH_ASSURANCE_SOURCE`

and **never** `CORROBORATED`, `PROVEN`, or `LEGALLY_ESTABLISHED`. The exception is a way to proceed honestly under a real constraint, not a way to launder a single source into corroboration.

10. Dissent (W5.3)

First-class state, four severities: `INFORMATIONAL`, `MINOR`, `MATERIAL`, `CRITICAL`.

Critical dissent covers integrity, provenance, source identity, acquisition legality, methodology, proof condition, neutrality, privilege, and replay.

Material unresolved dissent cannot disappear from the final finding (Article 11). It is not resolved by overruling it; it is carried.

11. Next best evidence (§21)

For each unresolved claim, candidate evidence is ranked:

$$\text{Priority} = \frac{\text{DiagnosticValue} \times \text{Independence} \times \text{Relevance} \times \text{Freshness} \times \text{AcquisitionFeasibility}}{\text{Cost} \times \text{Latency} \times \text{RightsRisk}}$$

Hard policy filters execute before optimization. A candidate that fails rights is removed from the set — it is never merely down-weighted into a low score that a sufficiently high diagnostic value could overcome. Article 4 is not a term in a ratio.

12. Implementation status

Component	Status
Reverse proof	<code>IMPLEMENTED</code> , <code>UNIT_TESTED</code>

Component	Status
Observability gate / observed absence	IMPLEMENTED, UNIT_TESTED, ADVERSARIAL_TESTED
Source independence	IMPLEMENTED, UNIT_TESTED, ADVERSARIAL_TESTED
Qualification states + single-source exception	IMPLEMENTED, UNIT_TESTED
Next best evidence	IMPLEMENTED, UNIT_TESTED
Contradiction matrix	IMPLEMENTED (pre-existing pkg/insurance/contradiction)
Trust calculus	IMPLEMENTED (pre-existing pkg/core/trustcalc)
Calibration	IMPLEMENTED (pre-existing pkg/governance/calibration)
Full EQF orchestration across all nine components	DESIGNED — the components exist and are individually tested; a single end-to-end orchestrator binding all nine is not built